

Emergent Management of Intracerebral Hemorrhage

By Santosh B. Murthy, MD, MPH, FNCS

REVIEW ARTICLE



CONTINUUM AUDIO
INTERVIEW AVAILABLE
ONLINE

ABSTRACT

OBJECTIVE: Nontraumatic intracerebral hemorrhage (ICH) is a potentially devastating cerebrovascular disorder. Several randomized trials have assessed interventions to improve ICH outcomes. This article summarizes some of the recent developments in the emergent medical and surgical management of acute ICH.

LATEST DEVELOPMENTS: Recent data have underscored the protracted course of recovery after ICH, particularly in patients with severe disability, cautioning against early nihilism and withholding of life-sustaining treatments. The treatment of ICH has undergone rapid evolution with the implementation of intensive blood pressure control, novel reversal strategies for coagulopathy, innovations in systems of care such as mobile stroke units for hyperacute ICH care, and the emergence of newer minimally invasive surgical approaches such as the endoport and endoscope-assisted evacuation techniques.

ESSENTIAL POINTS: This review discusses the current state of evidence in ICH and its implications for practice, using case illustrations to highlight some of the nuances involved in the management of acute ICH.

INTRODUCTION

Acute nontraumatic intracerebral hemorrhage (ICH) accounts for 10% to 15% of all strokes and is a neurologic emergency.¹ ICH is the most severe type of stroke with one-third of patients dying before hospital discharge and nearly one-half of surviving patients remaining severely disabled at 6 months.^{2,3} More than 3 million people experience an ICH every year worldwide, including 80,000 in the United States where the incidence of ICH is about 43 cases per 100,000 person-years.^{4,5} ICH occurs more frequently in men and the incidence increases with age.⁶ Population-based studies have reported a higher incidence of ICH in Black and Hispanic populations compared with White populations in the United States.^{6,7} However, over the past few decades, the incidence of ICH has increased by 11% in the United States, with trends strikingly higher among younger and middle-aged people and Black people.⁸ The overall 30-day case fatality rate of ICH appears to have decreased from 40% to 33% between 1985 and 2011, but the 1-year

CITE AS:

CONTINUUM (MINNEAPOLIS, MINN)
2024;30(3, NEUROCRITICAL CARE):
641–661.

Address correspondence to
Dr Santosh B. Murthy, 525 E 68th
St, Room F610, New York, NY 10065,
sam9200@med.cornell.edu.

RELATIONSHIP DISCLOSURE:

Dr Murthy has received personal compensation in the range of \$10,000 to \$49,999 for serving as an expert witness for stroke and neurologic disorders. The institution of Dr Murthy has received research support from the National Institutes of Health (NIH)/National Institute of Neurological Disorders and Stroke.

UNLABELED USE OF PRODUCTS/INVESTIGATIONAL USE DISCLOSURE:

Dr Murthy reports no disclosure.

© 2024 American Academy
of Neurology.

case fatality rate has remained steady at 50% and 55%.⁹ These trends are worrisome given the high morbidity and quality-adjusted life years lost.

Similar to ischemic stroke, ICH has several important risk factors that include older age, male sex, excessive alcohol consumption, and illicit drug use.¹⁰ Vasculopathy in the form of cerebral small vessel disease underlies most cases of ICH. One of the most common types of small vessel disease is deep perforator arteriopathy, often due to long-standing poorly controlled hypertension, which can lead to microaneurysm formation or lipohyalinosis, in turn resulting in ICH or lacunar infarction, respectively.¹¹ ICH due to deep perforator arteriopathy often involves the basal ganglia, thalamus, deep portions of the cerebellum, and brainstem.¹² Another form of cerebral small vessel disease is cerebral amyloid angiopathy, characterized by the deposition of amyloid- β protein in the cortical blood vessels, leading to lobar ICH.¹³ Notably, patients with amyloid angiopathy have an up to 9% risk of ICH recurrence.¹³ Regardless of the underlying pathophysiology, ICH is characterized by the primary injury that occurs from the hematoma followed by secondary injury mediated by inflammation and oxidative stress,¹⁴ which combine to result in neurologic deterioration and medical complications during the acute period of ICH (FIGURE 4-1). This article reviews the current state of knowledge in ICH and outlines ongoing investigations that are likely to shift the paradigm of acute ICH management in the future.

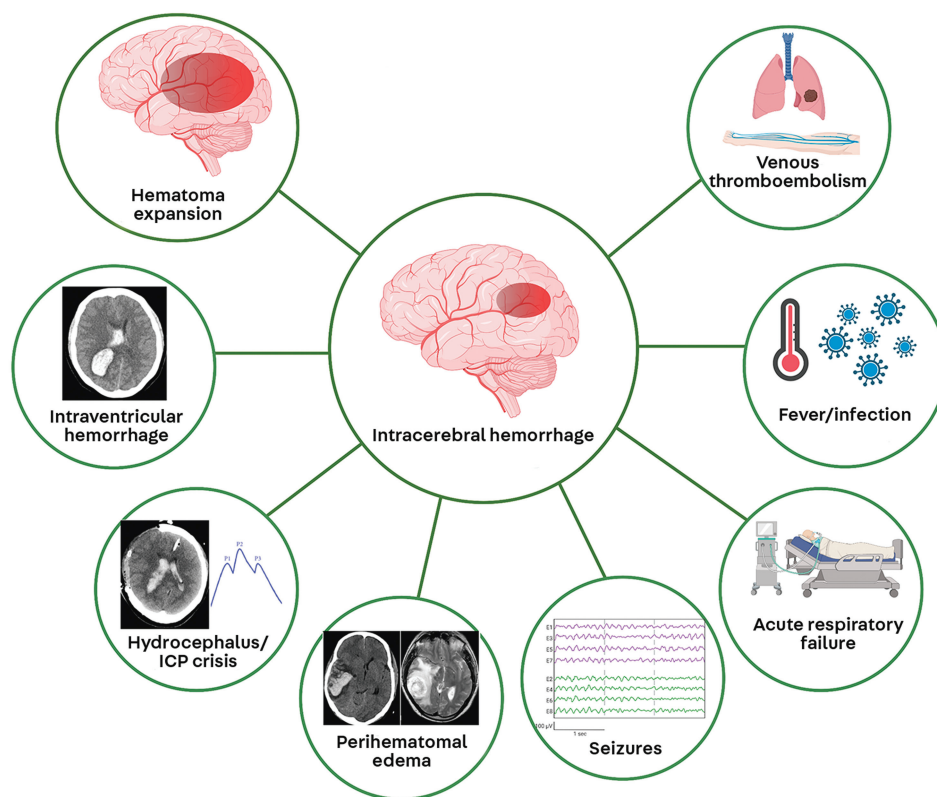


FIGURE 4-1

Summary of early complications after spontaneous intracerebral hemorrhage.

ICP = intracranial pressure.

HEMATOMA EXPANSION

Hematoma expansion is the interval increase in the volume of ICH between baseline and follow-up CT scans and occurs in about 20% to 25% of patients with ICH.¹⁵ Although hematoma expansion can evolve over 24 hours, most ICHs expand in the first 6 hours, with the peak expansion occurring in the first 3 hours after symptom onset.¹⁶ Observational data suggest that hematoma expansion is more common in deep ICH than lobar ICH and mediates the relationship between deep ICH location and poor outcomes.¹⁷ Factors associated with hematoma expansion include a shorter time from symptom onset to baseline imaging, higher ICH volume on baseline imaging, pre-ICH use of antithrombotic medications, and presence of the spot sign on CT angiography (CTA).¹⁶ Extravasation of iodinated contrast within the hematoma on CTA results in the spot sign and represents an actively bleeding vessel (FIGURE 4-2). As such, the CTA spot sign has a modest sensitivity of 57% and a false positive rate of 12% to accurately predict hematoma expansion.¹⁸ The sensitivity of this sign is limited because arterial-phase CTA images are sometimes procured before the contrast material has had sufficient time to reach the point of extravasation.¹⁹ Hematoma expansion is an independent predictor of poor outcomes.²⁰ In fact, every 1-mL increase in ICH volume increases the odds of death and major disability by 5%.²¹ Therefore, the concept of “time is brain,” widely used for ischemic stroke, also applies to ICH, and the emergent management of ICH centers on the prevention or mitigation of hematoma expansion. Three key interventions that potentially play a role in limiting hematoma expansion include acute blood pressure lowering, reversal of coagulopathy, and administration of hemostatic agents.¹⁰

Blood Pressure Control

The acute phase of ICH is often accompanied by sustained elevations in blood pressure (systolic blood pressure greater than 180 mm Hg), which is considered a

KEY POINTS

- Intracerebral hemorrhage (ICH) is characterized by the primary injury that occurs from the hematoma followed by secondary injury mediated by inflammation and oxidative stress.
- Factors associated with hematoma expansion include a shorter time from symptom onset to baseline imaging, higher ICH volume on baseline imaging, pre-ICH use of antithrombotic medications, and presence of the spot sign on CT angiography.
- Three key interventions that potentially play a role in limiting intracerebral hematoma expansion include acute blood pressure lowering, reversal of coagulopathy, and administration of hemostatic agents.

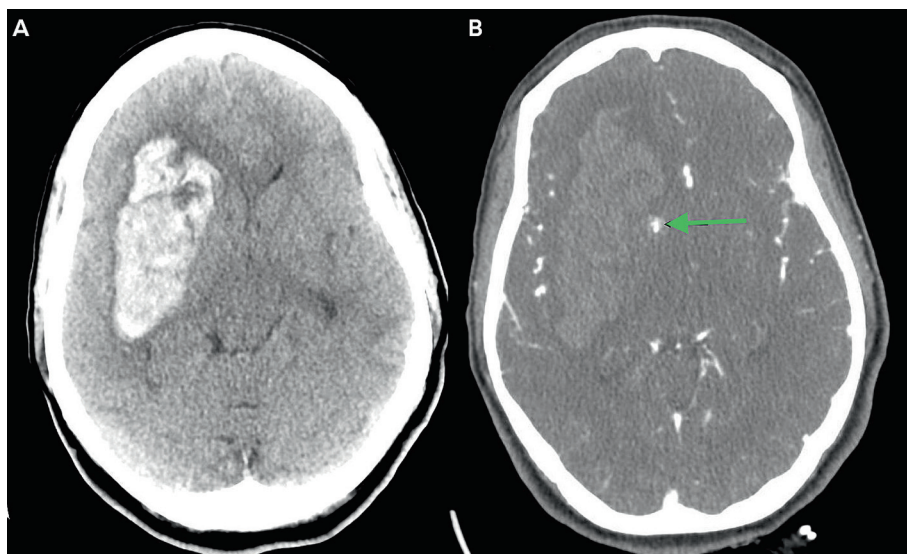


FIGURE 4-2

Acute neuroimaging marker of hematoma expansion. A CT scan of the head showing an acute intracerebral hemorrhage (A), with a corresponding CT angiography spot sign (B, arrow) indicating a high risk of hematoma expansion.

hypertensive emergency.²² Acute blood pressure elevation is independently associated with hematoma expansion and poor outcomes after ICH.²³ As a result, blood pressure control has been considered a potential therapeutic target for decades. Two randomized clinical trials have evaluated the safety and efficacy of acute blood pressure reduction in this setting. In INTERACT2 (Intensive Blood Pressure Reduction in Acute Cerebral Hemorrhage Trial), patients with ICH randomly assigned to intensive blood pressure lowering (systolic blood pressure less than 140 mm Hg) had a nonsignificant trend toward favorable functional outcomes compared with those in the liberal blood pressure control arm (systolic blood pressure 140 mm Hg to 180 mm Hg).²⁴

However, the ATACH-2 (Antihypertensive Treatment of Acute Cerebral Hemorrhage) trial did not show any benefit for the primary outcome of major disability or death among patients randomly assigned to the intensive blood pressure control (systolic blood pressure less than 140 mm Hg).²⁵ In the context of these negative trials, there is uncertainty about the optimal blood pressure goals after acute ICH. However, post hoc analyses of these trials have shed more light on the efficacy of blood pressure reduction. For instance, secondary analyses of the ATACH-2 trial demonstrated that intensive blood pressure control reduces hematoma expansion and improves functional outcomes in deep ICH and when initiated within 2 hours of symptom onset.^{26,27} Additionally, a pooled analysis of the INTERACT2 and ATACH-2 trials showed that fluctuations in blood pressure in the acute phase of ICH were associated with hematoma expansion, death, and disability.²⁸ Based on these results, the current American Heart Association guidelines recommend a systolic blood pressure target of 130 mm Hg to 150 mm Hg, particularly when the presenting systolic blood pressure is between 150 mm Hg and 220 mm Hg.²⁹ Further, avoiding peaks and variability in systolic blood pressure and initiating blood pressure lowering within 2 hours of symptom onset are also recommended.²⁹

Reversal of Coagulopathy

Approximately 30% of patients with ICH are on antiplatelet therapy, and about 20% are on anticoagulation medications.^{30,31} Recent trends suggest that the increasing use of antithrombotic medications may be partly responsible for the rising incidence of ICH in the United States.⁸ As mentioned previously, antithrombotic therapy is an independent predictor of hematoma expansion.¹⁶ Although studies evaluating the relationship between antiplatelet therapy preceding the ICH and functional outcomes have yielded conflicting results, anticoagulation has consistently been found to be associated with poor ICH outcomes.³¹⁻³³ Furthermore, among anticoagulation subtypes, patients with ICH who were previously on oral factor Xa inhibitors have better functional outcomes at discharge than those with warfarin-related ICH, but factor Xa inhibitor-associated ICH is associated with poor outcomes compared with no anticoagulation therapy.³¹ Therefore, immediate discontinuation of the antithrombotic agent and emergent correction of coagulopathy are warranted.²⁹

Historically, platelet transfusions have been the mainstay for ICH with concomitant antiplatelet therapy. In the PATCH (Platelet Transfusion in Cerebral Haemorrhage) trial, platelet transfusions were associated with major disability and death when compared with standard medical therapy in patients with a supratentorial ICH and recent use of antiplatelet therapy who did not have planned surgical intervention.³⁴ The current guidelines, therefore, do not

recommend the use of platelet transfusions for ICH in the setting of antiplatelet therapy except in patients who need emergent surgical intervention.²⁹ Another medication used in antiplatelet-related ICH is desmopressin, given its ability to potentially improve platelet function by increasing the release of von Willebrand factor and factor VIII from the endothelium.³⁵ However, the role of desmopressin in limiting or preventing hematoma expansion is unclear in patients with ICH who are on antiplatelet agents.³⁶

Specific reversal agents for coagulopathy due to anticoagulant medication use are highlighted in **TABLE 4-1**.^{29,37-39} Protamine is the reversal strategy of choice for unfractionated heparin and low-molecular-weight heparin, although it only partially reverses the latter.²⁹ The reversal of warfarin-associated coagulopathy entails the use of four-factor prothrombin complex concentrates, which have been shown to normalize the international normalized ratio significantly faster than fresh frozen plasma.³⁷ Prothrombin complex concentrates are ideally used in conjunction with vitamin K.³⁸ Coagulopathy resulting from dabigatran, a direct thrombin inhibitor, may be reversed by the specific reversal agent idarucizumab, an antibody fragment-binding agent that binds to dabigatran, rendering the medication inactive and neutralizing the anticoagulant effect.³⁸ If idarucizumab is not available, the intervention of choice is renal replacement therapy with or without administration of prothrombin complex concentrates.³² For ICHs associated with the use of direct factor Xa inhibitors, such as apixaban, rivaroxaban, and edoxaban, the use of either andexanet alfa or prothrombin complex concentrates is recommended.²⁹ Andexanet alfa, a modified recombinant inactive factor Xa designed to reverse factor Xa inhibitors, can also bind and inhibit the activity of tissue factor pathway inhibitor and can consequently increase thrombin generation and ultimately promote a procoagulant state.³⁹ In fact, in the ANNEXA-4 (Andexanet Alfa, a Novel Antidote to the Anticoagulation Effects of Factor Xa Inhibitors) study, the rate of incident thrombotic events was 10% in patients treated with andexanet alfa, but given this was a single-arm study, it was unclear if the intrinsic thrombotic risk posed by the ICH and the rebound hypercoagulability from stopping the antithrombotic medication also played a role.³⁹ The lack of head-to-head comparison trials notwithstanding, there do not appear to be discernable differences in outcomes between andexanet alfa and prothrombin complex concentrates based on observational data, but cost-effectiveness studies seem to favor the use of prothrombin complex concentrates.⁴⁰

Administration of Hemostatic Agents

Hematoma expansion has been the target of several therapeutic interventions, particularly hemostatic medications such as recombinant factor VIIa and tranexamic acid. The FAST (Factor VII for Acute Hemorrhagic Stroke Treatment) trial, in which patients with an ICH were randomly assigned to receive recombinant activated factor VII or placebo, showed that hemostatic therapy with recombinant activated VII reduced hematoma growth but did not improve survival or functional outcome.⁴¹ The safety and efficacy of tranexamic acid compared with placebo was investigated in the TICH-2 (Tranexamic acid for hyperacute primary IntraCerebral Haemorrhage) trial, which was neutral for the primary outcome of functional recovery at 90 days.⁴² However, benefits were observed for early death, hematoma expansion, and serious adverse events.⁴²

KEY POINTS

- The current American Heart Association guidelines recommend a systolic blood pressure target of 130 mm Hg to 150 mm Hg for patients with ICH, particularly when the presenting systolic blood pressure is between 150 mm Hg and 220 mm Hg.
- Current guidelines do not recommend the use of platelet transfusions for ICH in the setting of antiplatelet therapy except in patients who need emergent surgical intervention.
- The reversal of warfarin-associated coagulopathy in patients with ICH entails the use of four-factor prothrombin complex concentrates.
- Coagulopathy resulting from the use of dabigatran, a direct thrombin inhibitor, may be reversed in patients with ICH by the specific reversal agent idarucizumab, an antibody fragment-binding agent that binds to dabigatran, rendering the medication inactive and neutralizing the anticoagulant effect.
- For ICHs associated with the use of direct factor Xa inhibitors, such as apixaban, rivaroxaban, and edoxaban, the use of either andexanet alfa or prothrombin complex concentrates is recommended.

Hemostatic therapy has also been used in a subset of ICH deemed to be high risk for hematoma growth, such as in patients with a CTA spot sign. For instance, in parallel investigator-initiated trials, SPOTLIGHT (“Spot Sign” Selection of Intracerebral Hemorrhage to Guide Hemostatic Therapy) in Canada and STOP-IT (The Spot Sign for Predicting and Treating ICH Growth Study) in the United States, random assignment to recombinant activated factor VII did not result in improved radiographic or clinical outcomes.⁴³ A notable limitation of the trials was slow recruitment that led to underpowered studies. Moreover, hematoma expansion occurred in most cases between the baseline CT and the early postdose CT (performed within an hour of the study agent administration), limiting any potential treatment effect of hemostatic therapy.⁴⁴ These converging lines of evidence, therefore, do not support the administration of hemostatic drugs to improve functional recovery after ICH. In this regard, the ongoing FASTEST (Factor VIIa for Hemorrhagic Stroke Treatment at Earliest

TABLE 4-1 Reversal Strategies for Intracerebral Hemorrhage Related to Antithrombotic Medications

Antithrombotic agent	Recommended reversal strategy	Current evidence
Antiplatelet	Platelet transfusion only for emergent neurosurgical procedures after antiplatelet-related intracerebral hemorrhage (ICH) Unclear role of desmopressin in decreasing hematoma expansion	Platelet transfusions were associated with poor outcomes in patients with antiplatelet-related ICH managed conservatively in the PATCH (Platelet Transfusion in Cerebral Haemorrhage) trial ³⁴
Warfarin	International normalized ratio (INR) 1.3-1.9: four-factor prothrombin complex concentrate 10-20 IU/kg INR ≥2.0: four-factor prothrombin complex concentrate 25-50 IU/kg Vitamin K 10 mg IV	Prothrombin complex concentrate has a significantly faster time to normalization of INR than fresh frozen plasma (INCH [International Normalized Ratio Normalization in Coumadin Associated Intracerebral Haemorrhage] trial) ³⁷ ; vitamin K administration prevents rebound increase in INR ³⁷
Dabigatran	Activated charcoal if time from last dose <2 hours Idarucizumab is the treatment of choice If idarucizumab is not available, then four-factor prothrombin complex concentrate with or without renal replacement therapy	Idarucizumab is a monoclonal antibody fragment binding-agent that binds to dabigatran, rendering the medication inactive and neutralizing the anticoagulant effect ³⁸
Factor Xa inhibitors (apixaban, rivaroxaban, edoxaban)	Activated charcoal if time from last dose <2 hours Andexanet or four-factor prothrombin complex concentrate	Andexanet alfa is a modified recombinant inactive factor Xa that reverses factor Xa inhibitors ³⁹ ; no head-to-head comparison between andexanet alfa and four-factor prothrombin complex concentrates has been studied
Heparin (unfractionated and low-molecular-weight)	Protamine is the treatment of choice for the reversal of heparin-related coagulopathy	Protamine only partially reverses the effect of low molecular weight heparin ²⁹ ; the IV protamine infusion rate should not exceed 50 mg over 10 min because of the risk of hypotension and bronchoconstriction

Possible Time) trial will provide more insight into the ultra-early administration of hemostatic therapy because the enrollment window is within 2 hours of symptom onset.⁴⁵

INTRAVENTRICULAR HEMORRHAGE AND OBSTRUCTIVE HYDROCEPHALUS

Blood within the ventricles, intraventricular hemorrhage (IVH), is seen in up to one-half of all patients with ICH.²² Although IVH is present on the admission CT scan in most patients, the timing of IVH can vary and may be delayed in about 15% of cases.⁴⁶ Like the parenchymal hematoma, IVH can expand in about one-quarter of cases.⁴⁷ The presence and subsequent growth of IVH portend poor outcomes after ICH.^{46,47} Additionally, the IVH clot may lead to blockage of CSF circulation, and the ensuing obstructive hydrocephalus causes depressed level of consciousness, elevated intracranial pressure, and airway compromise.¹⁰ Therefore, an external ventricular drain (EVD) is recommended for patients with ICH and moderate to large IVH who have impaired consciousness to facilitate drainage of CSF and alleviate high intracranial pressure.²⁹

The role of intraventricular thrombolysis in expediting the clearance of IVH and relieving hydrocephalus was evaluated in the CLEAR III (Clot Lysis: Evaluating Accelerated Resolution of Intraventricular Hemorrhage Phase III) trial, in which patients with a large IVH and radiographic obstruction of the third or fourth ventricle or both ventricles who had an EVD were randomly assigned to receive intraventricular thrombolysis or saline.⁴⁸ The use of intraventricular thrombolysis improved survival but did not translate to better functional recovery. Of note, the median IVH removal was about 65% of the IVH volume, and subgroup analyses showed less disability with intraventricular thrombolysis in patients who had large IVH (greater than 20 mL) and those who had more than 80% of the IVH evacuated compared with patients who did not receive thrombolysis.⁴⁸ A subsequent individual patient-level meta-analysis showed better functional outcomes with the use of intraventricular thrombolysis, particularly when initiated within 48 hours of symptom onset.⁴⁹ Furthermore, intraventricular thrombolysis in conjunction with lumbar drainage of CSF may additionally help with obviating the need for permanent CSF diversion.⁵⁰ Taken together, it may be reasonable to consider intraventricular thrombolysis in patients with large IVH, obstructive hydrocephalus, and decreased consciousness, especially in the context of high intracranial pressure, keeping in mind that this intervention likely reduces mortality but may not affect functional outcomes.²⁹ **CASE 4-1** illustrates the hyperacute management of ICH to mitigate hematoma expansion, mass effect, and cerebral herniation.

PERIHEMATOMAL EDEMA AND MASS EFFECT

Perihematomal edema, or the swelling around the hematoma, is an imaging marker of secondary injury after ICH.⁵¹ Perihematomal edema usually starts as early as 4 to 6 hours after ICH, peaks in 3 to 7 days, and then plateaus, with the rate of perihematomal edema growth being the highest in the first 24 hours after ICH onset.⁵² Given that perihematomal edema contributes to early neurologic deterioration after ICH,⁵³ the temporal trend in the evolution of perihematomal edema offers an important guide for the period of close monitoring likely needed after ICH onset. Factors purported to play a role in perihematomal edema growth include admission blood pressure, initial hematoma volume, hematoma expansion, hyperglycemia, body temperature, and serum sodium

KEY POINTS

- Evidence does not support the administration of hemostatic drugs to improve functional recovery after ICH.
- An external ventricular drain is recommended for patients with ICH and moderate to large intraventricular hemorrhage who have impaired consciousness to facilitate drainage of CSF and alleviate high intracranial pressure.
- It may be reasonable to consider intraventricular thrombolysis in patients with large intraventricular hemorrhage, obstructive hydrocephalus, and decreased consciousness, especially in the context of high intracranial pressure.
- Prophylactic hyperosmolar therapy after ICH has a limited role, but it is a first-line treatment option for transiently reducing intracranial pressure or mitigating perihematomal edema once edema commences.
- The American Heart Association guidelines recommend craniotomy for ICH as a lifesaving option in the context of clinical deterioration but not necessarily for improving functional outcomes.

concentration.⁵¹ Perihematomal edema evolution in the first 72 hours after ICH is independently associated with functional disability after ICH, more so in deep ICH than lobar ICH.⁵⁴ Perihematomal edema expansion causes mass effect, midline shift, and cerebral and cerebellar herniation, which play a role in clinical worsening.⁵¹

The cornerstone of treatment of perihematomal edema is osmotherapy. Prophylactic hyperosmolar therapy after ICH has a limited role, but it is a first-line treatment option for transiently reducing intracranial pressure or mitigating perihematomal edema once edema commences.²⁹ A meta-analysis suggested the

CASE 4-1

A 60-year-old man presented with sudden-onset lethargy and right-sided hemiparesis. He had hypertension, type 2 diabetes mellitus, and nonvalvular atrial fibrillation for which he was on full-dose apixaban. His blood pressure on presentation was 174/82 mm Hg. Head CT without contrast showed a large left temporoparietal acute intracerebral hemorrhage (ICH) (FIGURE 4-3A). He was started on a nicardipine drip and given prothrombin complex concentrates and vitamin K to reverse the coagulopathy from apixaban. He was then transferred to a tertiary care facility with a neurocritical care unit. On arrival there, a repeat CT scan of his head showed the interval development of intraventricular hemorrhage (FIGURE 4-3B). An external ventricular drain was placed emergently (FIGURE 4-3C); however, he continued to have persistently elevated intracranial pressure with little response to osmotherapy. After discussions with the family, the decision for surgical hematoma evacuation was made. He underwent a decompressive hemicraniectomy with hematoma evacuation (FIGURE 4-3C and 4-3D). Over the next 2 weeks, tracheostomy and gastrostomy tubes were placed. At the time of discharge to rehabilitation, he was awake, was able to follow commands, and had significant right hemiparesis. At his 1-year follow-up clinic visit, he was able to ambulate independently.

COMMENT

This case highlights the importance of rapid blood pressure control and reversal of coagulopathy following diagnosis of ICH. An external ventricular drain is indicated in the context of intraventricular hemorrhage, hydrocephalus, or both. The current American Heart Association guidelines recommend considering surgery for supratentorial ICH in patients with large hematomas or neurologic deterioration as a lifesaving approach to reduce mortality although the effect on functional recovery remains uncertain at this time.²⁹

superiority of hypertonic saline over mannitol in obtaining sustained control of intracranial pressure.⁵⁵ Whether symptom-based bolus dosing or a targeted sodium level is the ideal strategy for hypertonic saline therapy remains a topic of debate.⁵⁶ Another intervention to potentially attenuate perihematoma edema expansion is intensive blood pressure control (systolic blood pressure less than 140 mm Hg) as observed in a secondary analysis of the ATACH-2 trial.⁵⁷ The emergence of minimally invasive surgery for ICH offers another potential option for preventing or limiting ICH-related secondary injury by facilitating early hematoma evacuation. In the MISTIE II trial (Minimally invasive surgery and

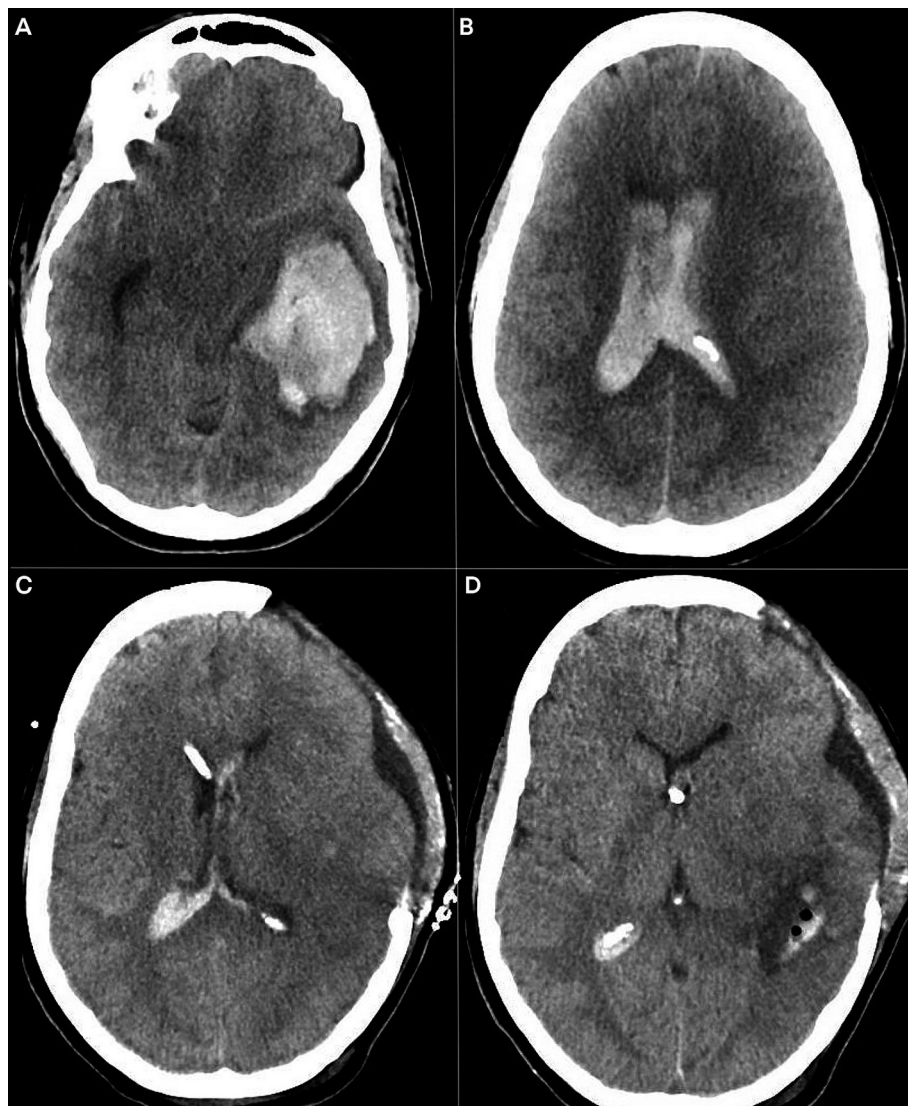


FIGURE 4-3

Imaging from the patient in **CASE 4-1**. A CT scan shows an acute intraparenchymal hemorrhage in the left temporal lobe (A) with concurrent intraventricular hemorrhage (B) and interval resolution of the burden of hematoma after surgical clot evacuation, decompressive hemicraniectomy, and placement of an external ventricular drain (C, D).

rt-PA [recombinant tissue-type plasminogen activator] in ICH evacuation phase II), hematoma evacuation with surgery, thrombolysis, or both, compared with the standard medical management, resulted in a significant reduction in perihematomal edema.⁵⁸ Lastly, hypothermia has been evaluated in several small observational studies, but its benefit in limiting perihematomal edema evolution is unclear.⁵⁹

SURGERY FOR INTRACEREBRAL HEMORRHAGE

One of the most common surgical interventions for ICH is the placement of an EVD, often performed bedside, to relieve hydrocephalus, monitor intracranial pressure, and drain IVH, as discussed previously. In the acute period of ICH, hematoma expansion, IVH, hydrocephalus, perihematomal edema, mass effect, and cerebral herniation could prompt consideration for surgical intervention. Open and minimally invasive surgical approaches have been evaluated in ICH and are summarized in [TABLE 4-2](#).^{29,60-65}

Supratentorial Intracerebral Hemorrhage

The surgical management of supratentorial ICH has evolved over the years and encompasses three main techniques, which are discussed in the following sections.

DECOMPRESSIVE CRANIECTOMY. This is considered in the setting of mass effect either from the hematoma or from perihematomal edema, and the ensuing clinical changes secondary to cerebral herniation and elevated intracranial pressure.⁶⁶ Decompressive craniectomy involves the removal of the fronto-temporo-occipital bone with durotomy and dural expansion, which can lower the intracranial pressure to 30% of the presurgical levels.⁶⁷ It may be performed with or without hematoma evacuation and is often performed as a lifesaving measure.⁶⁸ Although there may be a mortality benefit for decompressive craniectomy, whether it improves functional outcomes after ICH is uncertain.²⁹

CRANIOTOMY. Another surgical technique is open craniotomy for ICH, which includes corticectomy and hematoma evacuation.⁶⁶ This approach was first evaluated in STICH (Surgical Trial in Intracerebral Haemorrhage) in which patients with a supratentorial ICH were randomly assigned to early open craniotomy or conservative medical management.⁶⁰ The trial showed no difference in functional outcome or mortality, but prespecified subgroup analyses revealed a potential benefit when the ICH was less than 1 cm from the cortex.⁶⁰ This formed the premise for the STICH II trial, which specifically included superficial lobar ICHs but failed to show any benefit on poor outcomes.⁶¹ A subsequent individual patient-level meta-analysis of more than 2000 patients with ICH showed favorable functional outcomes when the surgery was performed in the first 8 hours after symptom onset, in patients 50 to 69 years old, and when their Glasgow Coma Scale scores at randomization were between 9 and 12.⁶⁹ Correspondingly, the American Heart Association guidelines recommend craniotomy for ICH as a lifesaving option in the context of clinical deterioration but not necessarily for improving functional outcomes.²⁹

MINIMALLY INVASIVE SURGERY. The neutral results of the craniotomy trials spurred the exploration of minimally invasive surgical options for hematoma evacuation.

Surgical technique	Description of the procedure	Impact on outcome
External ventricular drain (EVD)	An EVD is a flexible plastic catheter placed in the frontal horn of the lateral ventricle or third ventricle for CSF drainage and to allow for monitoring of intracranial pressure	An EVD is recommended in patients with intracerebral hemorrhage (ICH) and moderate to large intraventricular hemorrhage who have impaired consciousness to facilitate drainage of CSF and alleviate high intracranial pressure ²⁹
EVD with intraventricular fibrinolytic therapy	An EVD is placed, and fibrinolytic therapy (such as recombinant tissue-type plasminogen activator) is serially administered directly into the ventricles to facilitate clot resolution	Based on the CLEAR III (Clot Lysis: Evaluating Accelerated Resolution of Intraventricular Hemorrhage Phase III) trial, intraventricular thrombolysis may be reasonable in patients with large intraventricular hemorrhage, obstructive hydrocephalus, and decreased consciousness to reduce mortality, but it may not affect functional outcomes ²⁹
Decompressive hemicraniectomy	Involves the removal of the fronto-temporo-occipital bone with durotomy and dural expansion; can be done with or without hematoma evacuation	Lifesaving procedure to relieve mass effect and cerebral herniation; retrospective data suggest mortality benefit but no association with functional outcomes ²⁹
Open craniotomy	Open craniotomy includes corticectomy and hematoma evacuation	Evaluated in the STICH (Surgical Trial in Intracerebral Haemorrhage) 1 and 2 trials, which showed no benefit on functional recovery ^{60,61} ; now recommended as a lifesaving procedure
Stereotactic clot aspiration	Stereotactic clot aspiration with thrombolytic therapy delivered through a catheter in the hematoma cavity	Evaluated in the MISTIE III (Minimally invasive surgery and rt-PA [recombinant tissue-type plasminogen activator] in ICH evacuation phase III) trial, which did not meet the primary efficacy outcome of major disability or death but did show lower mortality with surgery ⁶²
Minimally invasive parafascicular surgery with an endoport device	An endoport consisting of a sheath and an obturator is introduced into the longest axis of the hematoma cavity through a small craniotomy and a trans-sulcal parafascicular approach; once the sheath is placed, the obturator is removed, and the clot is evacuated with the assistance of exoscopic visualization	Evaluated in the ENRICH (Early Minimally Invasive Removal of Intracerebral Hemorrhage) trial in which minimally invasive surgery within 24 hours resulted in improved functional outcomes short-term and long-term compared with conventional medical management in lobar ICH ⁶³
Minimally invasive surgery with endoscopic evacuation	Entails performing a small craniotomy with stereotactic introduction of a port or sheath to the long axis of the hematoma followed by evacuation with an endoscope that has an aspiration device	Ongoing clinical trials such as MIND (Artemis in the Removal of Intracerebral Hemorrhage) and INVEST (Minimally Invasive Endoscopic Surgical Treatment with Apollo/Artemis in Patients with Brain Hemorrhage) will shed light on the efficacy of this procedure ⁶⁴
Suboccipital craniectomy or craniectomy with evacuation	Performed for posterior fossa ICH, mainly cerebellar ICH	For cerebellar ICH volume of 12-15 mL, medical management may be reasonable, monitoring for obstructive hydrocephalus and mechanical brainstem compression For cerebellar ICH volume > 15 mL consider surgery for reduced mortality; effect on functional outcome is unclear ⁶⁵

One such technique, stereotactic clot aspiration with thrombolytic therapy delivered through a catheter in the hematoma cavity, was investigated in the MISTIE III trial, which did not meet the primary efficacy outcome of major disability or death but did show lower mortality with surgery.⁶² Since the MISTIE III trial, there has been a renewed interest in the role of minimally invasive surgery for ICH, which has led to the emergence of two newer approaches. In the first technique, an endoport, consisting of a sheath and an obturator, is introduced into the longest axis of the hematoma cavity through a small craniotomy and a trans-sulcal parafascicular approach to minimize damage to the unaffected cerebral cortex.⁷⁰ Once the sheath is placed, the obturator is removed, and the clot is evacuated with the assistance of exoscopic visualization with conventional microsurgical techniques.⁷⁰ **CASE 4-2** illustrates the role of minimally invasive surgery for ICH in relieving mass effect and midline shift. This technique was investigated in the ENRICH (Early Minimally Invasive Removal of Intracerebral Hemorrhage) trial in which patients who underwent minimally invasive surgery within 24 hours of symptom onset for lobar ICH ranging in size from 30 mL to 80 mL had better functional recovery short-term and long-term than those who received standard medical treatment alone.⁶³ Additionally, there was a nearly 50% reduction in 30-day mortality in the surgery group compared with the control group (9.3% versus 18.0%).

The second minimally invasive method is endoscope-assisted evacuation, which requires a small craniotomy with stereotactic introduction of a port or sheath to the long axis of the hematoma followed by evacuation with an endoscope that has an aspiration device.⁷¹ The endoscopic techniques are being studied in the MIND (Artemis in the Removal of Intracerebral Hemorrhage) and

CASE 4-2

A 55-year-old man presented to the emergency department with acute-onset left hemiparesis and facial droop. He had a long-standing history of poorly controlled hypertension and was noted to have a blood pressure of 215/109 mm Hg. He reported no antithrombotic medication use. Head CT showed a large right-sided basal ganglia intracerebral hemorrhage with a 5-mm midline shift to the left. A nicardipine drip was started immediately, and he was transferred to the neurocritical care unit. The next morning, he was noted to be profoundly lethargic and was emergently intubated. A repeat head CT scan showed worsening edema and a 10-mm midline shift to the left (**FIGURE 4-4A**). Following discussions with his surrogate decision maker, he underwent minimally invasive surgery with the endoport device (**FIGURE 4-4B**) and postoperatively had significant improvement in midline shift and minimal residual hematoma (**FIGURE 4-4C**). He returned to the clinic after 6 months and was walking with a cane.

COMMENT

This case exemplifies the role of minimally invasive surgery as a lifesaving approach for supratentorial intracerebral hemorrhage in patients with large hematomas or neurologic deterioration, although the effect on functional recovery remains uncertain.²⁹

INVEST (Minimally Invasive Endoscopic Surgical Treatment with Apollo/Artemis in Patients with Brain Hemorrhage) clinical trials, which are underway.⁶⁴ A meta-analysis of randomized clinical trials found that minimally invasive surgery decreased the odds of poor outcomes by more than 50%.⁷² Moreover, time to minimally invasive surgery appears to be of vital importance in influencing functional recovery, in that every 1-hour delay decreases the odds of achieving favorable functional outcomes by 5%,⁷³ further emphasizing the “time is brain” principle in ICH.

Infratentorial Intracerebral Hemorrhage

Cerebellar ICH and the resultant perihematomal edema frequently result in obstructive hydrocephalus and mechanical brainstem compression and lead to early neurologic deterioration.¹⁴ Hence, early decompressive surgery with a suboccipital craniotomy is recommended, particularly when the hemorrhage volume is at least 15 mL (greater than 3 cm in diameter), or in the setting of early neurologic deterioration.²⁹ An individual patient-level meta-analysis found that there was no difference in outcomes between surgically and medically treated patients with cerebellar ICH.⁶⁵ A cutoff ICH volume of 12 mL to 15 mL was identified as a threshold below which the likelihood of benefit from surgery was low, whereas surgery for cerebellar ICH greater than 15 mL reduced mortality but did not affect functional outcome.⁶⁵ Although an EVD is placed in many cases at the time of surgery to relieve the obstructive hydrocephalus and determine subsequent cerebrospinal shunt needs, the use of an EVD alone without surgery is controversial given the theoretical risk of exacerbating upward cerebellar herniation.⁷⁴ **CASE 4-3** illustrates the

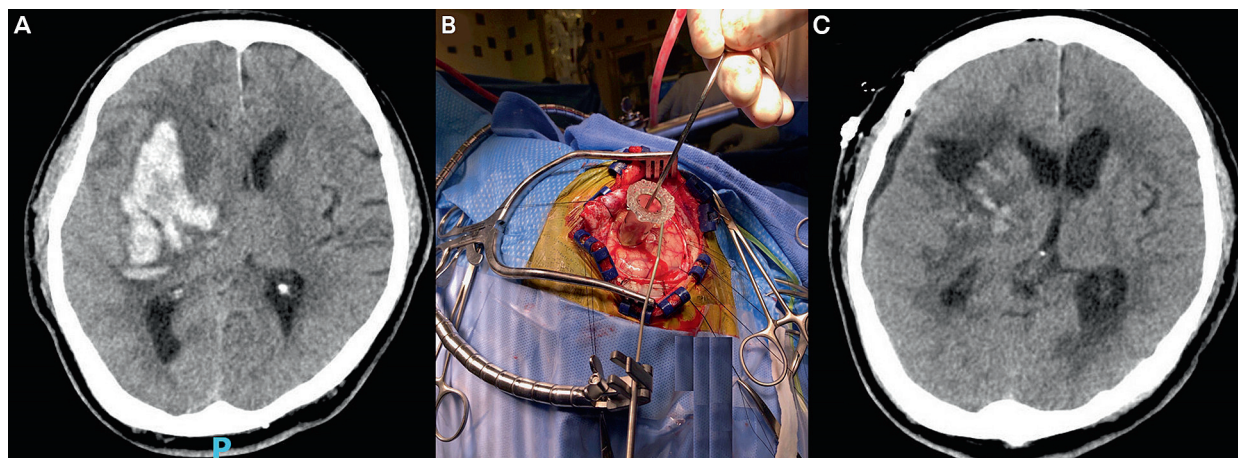


FIGURE 4-4

Imaging and photo from the patient in **CASE 4-2**. A CT scan shows an acute intracerebral hemorrhage with leftward midline shift and cerebellar herniation (A). After minimally invasive surgery with the endoport device (B), a repeat CT scan shows interval near-complete resolution of the hematoma (C).

conservative management of posterior fossa ICH in patients with an ICH volume of 12 mL to 15 mL.

SEIZURES

The frequency of new-onset seizures after ICH ranges from 5% to 10%.¹⁰ Although most seizures occur in the first week after ICH onset, prophylactic use of antiseizure medications is not recommended given their sedative effects, propensity for long-term adverse cognitive effects, and lack of an association with improved functional recovery after ICH.⁷⁵ It is recommended that antiseizure medications not be used prophylactically after ICH but rather to treat clinical or electrographic seizures.²⁹ It is also important to consider the use of continuous EEG monitoring for more than 24 hours in patients with ICH who have a

CASE 4-3

A 52-year-old woman presented to the emergency department with vertigo and gait imbalance. She was awake and did not have any weakness or sensory symptoms. Her medical history was notable for hypertension for which she was on three antihypertensive medications. A CT scan of her head without contrast showed an acute intracerebral hemorrhage with a volume of about 13 mL in the right deep cerebellar hemisphere (FIGURE 4-5A). There was partial effacement of the fourth ventricle without evidence of hydrocephalus. Her blood pressure on presentation was 188/104 mm Hg. She was started on a nicardipine drip for intensive blood pressure control. Based on her reassuring neurologic examination and absence of hydrocephalus on imaging, the decision was made to manage her medically and defer surgery. She was admitted to the neurocritical care unit for close monitoring. CT scan of her head on day 3 showed the evolution of perihematomal edema with partial effacement of the fourth ventricle (FIGURE 4-5B). There was, however, no evidence of hydrocephalus on imaging. She was started on hyperosmolar therapy for the edema, with a sodium concentration goal of greater than 145 mEq/L. Over the next few days, her condition remained stable, and she was eventually discharged to a rehabilitation facility. After 3 months, she was able to ambulate independently and did not have any gait imbalance or dysmetria.

COMMENT

This case outlines the management of posterior fossa hemorrhage. Historically, a hematoma volume of 15 mL or greater (3 cm in diameter) was the standard cutoff beyond which surgical decompression with a suboccipital craniectomy with or without external ventricular drain placement was recommended to reduce mortality.²⁹ In light of a recent individual patient-level meta-analysis, it may be reasonable to opt for conservative medical management and monitor patients closely for signs of neurologic deterioration or evidence of brainstem compression or hydrocephalus, particularly if the hematoma volume is between 12 mL and 15 mL and patients have minimal or no impairment of consciousness and a reassuring neurologic examination.⁶⁵

fluctuating mental status or when the neurologic examination findings are not explained by the radiographic characteristics of the bleed.²⁹ In a 2022 clinical trial, 50 patients with ICH were randomly assigned to receive either antiseizure medication (levetiracetam) or placebo and had continuous EEG initiated within 24 hours and recorded for at least 48 hours.⁷⁶ Treatment assignment to levetiracetam resulted in a significant reduction in a new clinical or electrographic seizure compared with the placebo group (19% versus 43%), attesting to both the high incidence of seizures after ICH and the relative safety and possible efficacy of levetiracetam in preventing seizures in the acute period of ICH and supporting the need for larger clinical trials.⁷⁶ The risk of seizures is not increased in the acute period alone. In fact, patients with ICH also experience a heightened long-term risk of seizures, which is the highest among stroke subtypes.⁷⁷ The long-term risk

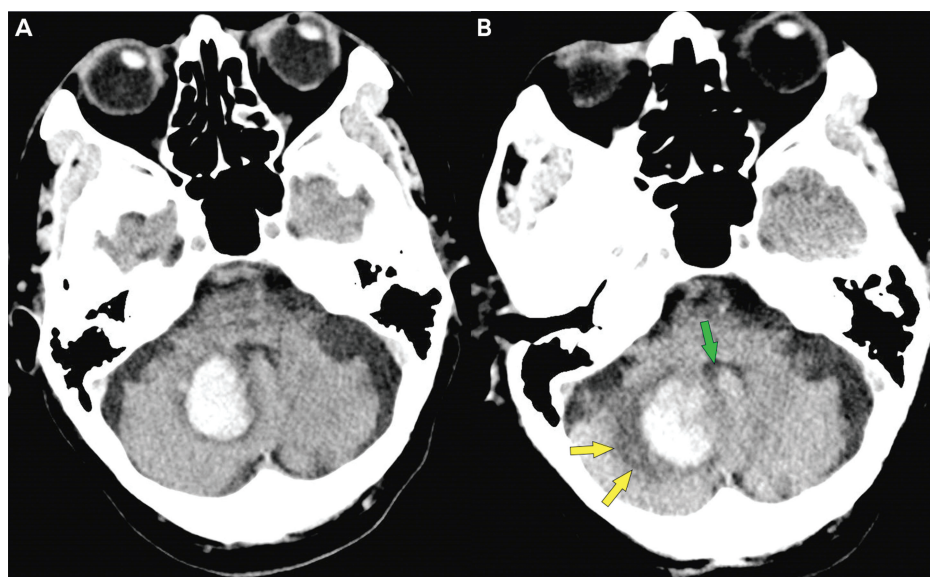


FIGURE 4-5

Imaging from the patient in **CASE 4-3**. An acute hematoma located in the right cerebellar hemisphere is seen on the CT scan with mild surrounding edema and partial obliteration of the fourth ventricle (A). An interval increase in the perihematomal edema (B, yellow arrows) is seen on repeat CT scan on day 3, with partial compression of the fourth ventricle (B, green arrow).

of seizure after ICH may be quantified by using the CAVE score, where 1 point is awarded for cortical location, young age (younger than 65 years), small hematoma volume, and early seizures (within 7 days of ICH).⁷⁸

ACUTE MEDICAL MANAGEMENT

The early complications of ICH such as mass effect and cerebral herniation from hematoma expansion and perihematomal edema, obstructive hydrocephalus, and seizures can cause progressive neurologic deterioration and respiratory compromise. With these potential complications in mind, adequate airway protection should be considered in patients without adequate gas exchange or a stable hemodynamic profile not just for those already admitted to the intensive care unit but also for those requiring transfer to another hospital.²⁹ Additionally, a Glasgow Coma Scale score of 8 or less, a worsening neurologic examination, or concerns about airway protection from ongoing neurologic injury may serve as guides to consider intubation and mechanical ventilation.²⁹

Elevated temperature and fever occur in about one-third of patients with ICH and may be driven by the primary brain injury and the accompanying systemic inflammatory response.⁷⁹ Fever is independently associated with poor outcomes after ICH⁸⁰; therefore, one goal of acute ICH management is to treat fever with medication and, if needed, with nonpharmacologic interventions such as surface or intravascular cooling devices. Although fever prevention may improve outcomes, the converse of reducing temperature in the absence of fever is not necessarily true; data are divided on the utility of therapeutic hypothermia in improving ICH outcome, although there may be a beneficial effect on perihematomal edema evolution.⁷⁹ The acute management of ICH also entails close monitoring of serum glucose levels because hypoglycemia (glucose less than 40 mg/dL to 60 mg/dL) and hyperglycemia (glucose greater than 180 mg/dL to 200 mg/dL) are associated with poor outcomes.^{81,82} In the NICE-SUGAR (Normoglycemia in Intensive Care Evaluation and Surviving Using Glucose Algorithm Regulation) trial, intensive blood glucose control (81 mg/dL to 100 mg/dL) resulted in high 30-day mortality in critically ill patients as compared with liberal glucose control (less than 180 mg/dL).⁸³ Furthermore, among patients with an ischemic stroke and hyperglycemia enrolled in the SHINE (Stroke Hyperglycemia Insulin Network Effort) trial, intensive blood glucose control (80 mg/dL to 130 mg/dL) failed to improve outcomes compared with standard glucose control (80 mg/dL to 179 mg/dL).⁸⁴ Extrapolating these results to the management of patients with ICH, targeting liberal glucose control while preventing extremes of glycemia (hypoglycemia and hyperglycemia) is recommended.²⁹

Venous thromboembolism (VTE), including deep venous thrombosis and pulmonary embolism, are serious medical complications encountered during hospitalization for ICH. The overall rate of symptomatic VTE after ICH ranges from 2% to 5%, with the incidence rising to nearly 25% with the incorporation of routine screening.⁸⁵⁻⁸⁷ Moreover, incident VTE is twofold to threefold higher after ICH than ischemic stroke.⁸⁸ Hence, prevention of VTE in patients with ICH is an important aspect of acute management, which may be achieved by a two-pronged approach. The first step involves the initiation of pneumatic compression devices on admission, which is supported by the CLOTS 3 (Clots in Legs Or sTockings after Stroke 3) trial, in which the use of these devices led to a reduction in the risk of VTE and 6-month mortality in 2876 patients with acute stroke, including 376 with ICH.⁸⁹ Second, chemoprophylaxis with low-dose

heparin or low-molecular-weight heparin compounds may be started 24 to 48 hours after ICH diagnosis and after a patient demonstrates hematoma stability.²⁹

PROGNOSTICATION AFTER INTRACRANIAL HEMORRHAGE

The mortality and morbidity associated with ICH have led clinicians to use prognostic scales to identify the subset of patients with ICH who have a high likelihood of poor outcomes.^{10,90-96} Over the years, this approach may have resulted in premature withdrawal of life-sustaining treatments leading to death in patients who might otherwise have survived, potentiating a self-fulfilling prophecy.²² Improvements in neurologic critical care and increasing awareness about the imperfect predictive accuracy of existing prognostic scales have led clinicians to be more hesitant about early withdrawal of life support. Furthermore, emerging data have shown that a considerable proportion (40%) of patients with ICH who have severe disability at 30 days experience significant recovery and have favorable outcomes at 1 year.⁹⁷ Therefore, recovery after ICH is protracted, warranting caution against early nihilism and withdrawal of life support. It is, therefore, prudent to postpone neuroprognostication in the first few days after ICH and integrate a shared decision-making model between the patient or surrogates and clinicians to ensure that the management plan aligns with the patient's wishes.²⁹

CONCLUSION

The landscape of ICH management continues to undergo rapid change. Advances in our understanding of intensive blood pressure reduction, outcome prognostication, and long-term neurologic recovery, the emergence of novel reversal agents for coagulopathy, a renewed interest in minimally invasive surgical techniques, and an emphasis on "time is brain" in ICH are likely to move the field forward and change the current paradigm of management of ICH.

REFERENCES

- 1 Tsao CW, Aday AW, Almarazooq ZI, et al. Heart disease and stroke statistics-2022 update: a report from the American Heart Association. *Circulation* 2022;145(8):e153-e639. doi:10.1161/CIR.0000000000001052
- 2 Fernando SM, Qureshi D, Talarico R, et al. Intracerebral hemorrhage incidence, mortality, and association with oral anticoagulation use: a population study. *Stroke* 2021;52(5):1673-1681. doi:10.1161/STROKEAHA.120.032550
- 3 van Asch CJ, Luitse MJ, Rinkel GJ, et al. Incidence, case fatality, and functional outcome of intracerebral haemorrhage over time, according to age, sex, and ethnic origin: a systematic review and meta-analysis. *Lancet Neurol* 2010;9(2):167-176. doi:10.1016/S1474-4422(09)70340-0
- 4 GBD 2019 Stroke Collaborators. Global, regional, and national burden of stroke and its risk factors, 1990-2019: a systematic analysis for the Global Burden of Disease Study 2019. *Lancet Neurol* 2021;20(10):795-820. doi:10.1016/S1474-4422(21)00252-0
- 5 Lioutas VA, Beiser AS, Aparicio HJ, et al. Assessment of incidence and risk factors of intracerebral hemorrhage among participants in the Framingham Heart Study between 1948 and 2016. *JAMA Neurol* 2020;77(10):1252-1260. doi:10.1001/jamaneurol.2020.1512
- 6 Labovitz DL, Halim A, Boden-Albala B, Hauser WA, Sacco RL. The incidence of deep and lobar intracerebral hemorrhage in whites, blacks, and Hispanics. *Neurology* 2005;65(4):518-522. doi:10.1212/01.wnl.0000172915.71933.00
- 7 Broderick JP, Brott T, Tomsick T, Huster G, Miller R. The risk of subarachnoid and intracerebral hemorrhages in blacks as compared with whites. *N Engl J Med* 1992;326(11):733-736. doi:10.1056/NEJM199203123261103
- 8 Bako AT, Pan A, Potter T, et al. Contemporary trends in the nationwide incidence of primary intracerebral hemorrhage. *Stroke* 2022;53(3):e70-e74. doi:10.1161/STROKEAHA.121.037332

KEY POINTS

- Time to minimally invasive surgery appears to be of vital importance in influencing functional recovery for patients with ICH, in that every 1-hour delay decreases the odds of achieving favorable functional outcomes by 5%.
- It is recommended that antiseizure medications not be used prophylactically after ICH but rather to treat clinical or electrographic seizures.
- The long-term risk of seizure after ICH may be quantified by using the CAFE score, where 1 point is awarded for cortical location, young age (younger than 65 years), small hematoma volume, and early seizures (within 7 days of ICH).
- Targeting liberal glucose control (less than 180 mg/dL) while preventing extremes of glycemia (hypoglycemia and hyperglycemia) is recommended for patients with ICH.

● Venous thromboembolism chemoprophylaxis with low-dose heparin or low-molecular-weight heparin compounds may be started 24 to 48 hours after ICH diagnosis and after a patient demonstrates hematoma stability.

● Recovery after ICH is protracted, warranting caution against early nihilism and withdrawal of life support.

- 9 Jolink WMT, Klijn CJM, Brouwers PJAM, Kappelle LJ, Vaartjes I. Time trends in incidence, case fatality, and mortality of intracerebral hemorrhage. *Neurology* 2015;85(15):1318–1324. doi:10.1212/WNL.0000000000002015
- 10 Cordonnier C, Demchuk A, Ziai W, Anderson CS. Intracerebral haemorrhage: current approaches to acute management. *Lancet* 2018;392(10154):1257–1268. doi:10.1016/S0140-6736(18)31878-6
- 11 van Asch CJJ, Velthuis BK, Rinkel GJE, et al. Diagnostic yield and accuracy of CT angiography, MR angiography, and digital subtraction angiography for detection of macrovascular causes of intracerebral haemorrhage: prospective, multicentre cohort study. *BMJ* 2015;351:h5762. doi:10.1136/bmj.h5762
- 12 Puy L, Parry-Jones AR, Sandset EC, et al. Intracerebral haemorrhage. *Nat Rev Dis Primers* 2023;9(1):14. doi:10.1038/s41572-023-00424-7
- 13 Charidimou A, Boulouis G, Gurol ME, et al. Emerging concepts in sporadic cerebral amyloid angiopathy. *Brain* 2017;140(7):1829–1850. doi:10.1093/brain/awx047
- 14 Magid-Bernstein J, Girard R, Polster S, et al. Cerebral hemorrhage: pathophysiology, treatment, and future directions. *Circ Res* 2022;130(8):1204–1229. doi:10.1161/CIRCRESAHA.121.319949
- 15 Morotti A, Boulouis G, Dowlatshahi D, et al. Intracerebral haemorrhage expansion: definitions, predictors, and prevention. *Lancet Neurol* 2023;22(2):159–171. doi:10.1016/S1474-4422(22)00338-6
- 16 Al-Shahi Salman R, Frantzas J, Lee RJ, et al. Absolute risk and predictors of the growth of acute spontaneous intracerebral haemorrhage: a systematic review and meta-analysis of individual patient data. *Lancet Neurol* 2018;17(10):885–894. doi:10.1016/S1474-4422(18)30253-9
- 17 Roh D, Boehme A, Young C, et al. Hematoma expansion is more frequent in deep than lobar intracerebral hemorrhage. *Neurology* 2020;95(24):e3386–e3393. doi:10.1212/WNL.00000000000010990
- 18 Phan TG, Krishnadas N, Lai VWY, et al. Meta-analysis of accuracy of the spot sign for predicting hematoma growth and clinical outcomes. *Stroke* 2019;50(8):2030–2036. doi:10.1161/STROKEAHA.118.024347
- 19 Rodríguez-Luna D, Dowlatshahi D, Aviv RI, et al. Venous phase of computed tomography angiography increases spot sign detection, but intracerebral hemorrhage expansion is greater in spot signs detected in arterial phase. *Stroke* 2014;45(3):734–739. doi:10.1161/STROKEAHA.113.003007
- 20 Dowlatshahi D, Demchuk AM, Flaherty ML, et al. Defining hematoma expansion in intracerebral hemorrhage: relationship with patient outcomes. *Neurology* 2011;76(14):1238–1244. doi:10.1212/WNL.0b013e3182143317
- 21 Delcourt C, Huang Y, Arima H, et al. Hematoma growth and outcomes in intracerebral hemorrhage: the INTERACT1 study. *Neurology* 2012;79(4):314–319. doi:10.1212/WNL.0b013e318260cbba
- 22 Sheth KN. Spontaneous intracerebral hemorrhage. *N Engl J Med* 2022;387(17):1589–1596. doi:10.1056/NEJMr2201449
- 23 Mullen MT, Anderson CS. Review of long-term blood pressure control after intracerebral hemorrhage: challenges and opportunities. *Stroke* 2022;53(7):2142–2151. doi:10.1161/STROKEAHA.121.036885
- 24 Anderson CS, Heeley E, Huang Y, et al. Rapid blood-pressure lowering in patients with acute intracerebral hemorrhage. *N Engl J Med* 2013;368(25):2355–2365. doi:10.1056/NEJMoal214609
- 25 Qureshi AI, Palesch YY, Barsan WG, et al. Intensive blood-pressure lowering in patients with acute cerebral hemorrhage. *N Engl J Med* 2016;375(11):1033–1043. doi:10.1056/NEJMoal603460
- 26 Leasure AC, Qureshi AI, Murthy SB, et al. Association of intensive blood pressure reduction with risk of hematoma expansion in patients with deep intracerebral hemorrhage. *JAMA Neurol* 2019;76(8):949–955. doi:10.1001/jamaneurol.2019.1141
- 27 Li Q, Warren AD, Qureshi AI, et al. Ultra-early blood pressure reduction attenuates hematoma growth and improves outcome in intracerebral hemorrhage. *Ann Neurol* 2020;88(2):388–395. doi:10.1002/ana.25793
- 28 Moullaali TJ, Wang X, Martin RH, et al. Blood pressure control and clinical outcomes in acute intracerebral haemorrhage: a preplanned pooled analysis of individual participant data. *Lancet Neurol* 2019;18(9):857–864. doi:10.1016/S1474-4422(19)30196-6
- 29 Greenberg SM, Ziai WC, Cordonnier C, et al. 2022 guideline for the management of patients with spontaneous intracerebral hemorrhage: a guideline from the American Heart Association/American Stroke Association. *Stroke* 2022;53(7):e282–e361. doi:10.1161/STR.0000000000000407
- 30 Khan NI, Siddiqui FM, Goldstein JN, et al. Association between previous use of antiplatelet therapy and intracerebral hemorrhage outcomes. *Stroke* 2017;48(7):1810–1817. doi:10.1161/STROKEAHA.117.016290
- 31 Xian Y, Zhang S, Inohara T, et al. Clinical characteristics and outcomes associated with oral anticoagulant use among patients hospitalized with intracerebral hemorrhage. *JAMA Netw Open* 2021;4(2):e2037438. doi:10.1001/jamanetworkopen.2020.37438
- 32 RESTART Collaboration. Effects of antiplatelet therapy after stroke due to intracerebral haemorrhage (RESTART): a randomised, open-label trial. *Lancet* 2019;393(10191):2613–2623. doi:10.1016/S0140-6736(19)30840-2

- 33 Law ZK, Desborough M, Roberts I, et al. Outcomes in antiplatelet-associated intracerebral hemorrhage in the tich-2 randomized controlled trial. *J Am Heart Assoc* 2021;10(5):e019130. doi:10.1161/JAHA.120.019130
- 34 Baharoglu MI, Cordonnier C, Al-Shahi Salman R, et al. Platelet transfusion versus standard care after acute stroke due to spontaneous cerebral haemorrhage associated with antiplatelet therapy (PATCH): a randomised, open-label, phase 3 trial. *Lancet* 2016;387(10038):2605–2613. doi:10.1016/S0140-6736(16)30392-0
- 35 Swieringa F, Lancé MD, Fuchs B, et al. Desmopressin treatment improves platelet function under flow in patients with postoperative bleeding. *J Thromb Haemost* 2015; 13(8):1503–1513. doi:10.1111/jth.13007
- 36 Mengel A, Stefanou MI, Hadaschik KA, et al. Early administration of desmopressin and platelet transfusion for reducing hematoma expansion in patients with acute antiplatelet therapy associated intracerebral hemorrhage. *Crit Care Med* 2020;48(7):1009–1017. doi:10.1097/CCM.0000000000004348
- 37 Steiner T, Poli S, Griebel M, et al. Fresh frozen plasma versus prothrombin complex concentrate in patients with intracranial haemorrhage related to vitamin K antagonists (INCH): a randomised trial. *Lancet Neurol* 2016; 15(6):566–573. doi:10.1016/S1474-4422(16)00110-1
- 38 Frontera JA, Lewin JJ, Rabinstein AA, et al. Guideline for reversal of antithrombotics in intracranial hemorrhage: a statement for healthcare professionals from the neurocritical care society and society of critical care medicine. *Neurocrit Care* 2016;24(1):6–46. doi:10.1007/s12028-015-0222-x
- 39 Milling TJ, Middeldorp S, Xu L, et al. Final study report of andexanet alfa for major bleeding with factor Xa inhibitors. *Circulation* 2023;147(13): 1026–1038. doi:10.1161/CIRCULATIONAHA.121.057844
- 40 Micieli A, Demchuk AM, Wijeyesundera HC. Economic evaluation of andexanet versus prothrombin complex concentrate for reversal of factor Xa-associated intracranial hemorrhage. *Stroke* 2021;52(4):1390–1397. doi:10.1161/STROKEAHA.120.031108
- 41 Kuohn LR, Witsch J, Steiner T, et al. Early deterioration, hematoma expansion, and outcomes in deep versus lobar intracerebral hemorrhage: the FAST trial. *Stroke* 2022;53(8): 2441–2448. doi:10.1161/STROKEAHA.121.037974
- 42 Sprigg N, Flaherty K, Appleton JP, et al. Tranexamic acid for hyperacute primary Intracerebral Haemorrhage (TICH-2): an international randomised, placebo-controlled, phase 3 superiority trial. *Lancet* 2018;391(10135): 2107–2115. doi:10.1016/S0140-6736(18)31033-X
- 43 Gladstone DJ, Aviv RI, Demchuk AM, et al. Effect of recombinant activated coagulation factor vii on hemorrhage expansion among patients with spot sign-positive acute intracerebral hemorrhage: the SPOTLIGHT and STOP-IT randomized clinical trials. *JAMA Neurol* 2019; 76(12):1493–1501. doi:10.1001/jamaneurol.2019.2636
- 44 Al-Ajlan FS, Gladstone DJ, Song D, et al. Time course of early hematoma expansion in acute spot-sign positive intracerebral hemorrhage: prespecified analysis of the SPOTLIGHT randomized clinical trial. *Stroke* 2023;54(3): 715–721. doi:10.1161/STROKEAHA.121.038475
- 45 Naidech AM, Grotta J, Elm J, et al. Recombinant factor VIIa for hemorrhagic stroke treatment at earliest possible time (FASTEST): protocol for a phase III, double-blind, randomized, placebo-controlled trial. *Int J Stroke* 2022;17(7):806–809. doi:10.1177/17474930211042700
- 46 Witsch J, Bruce E, Meyers E, et al. Intraventricular hemorrhage expansion in patients with spontaneous intracerebral hemorrhage. *Neurology* 2015;84(10):989–994. doi:10.1212/WNL.0000000000001344
- 47 Roh DJ, Asonye IS, Carvalho Poyraz F, et al. Intraventricular Hemorrhage Expansion in the CLEAR III Trial: a post hoc exploratory analysis. *Stroke* 2022;53(6):1847–1853. doi:10.1161/STROKEAHA.121.037438
- 48 Hanley DF, Lane K, McBee N, et al. Thrombolytic removal of intraventricular haemorrhage in treatment of severe stroke: results of the randomised, multicentre, multiregion, placebo-controlled CLEAR III trial. *Lancet* 2017;389(10069): 603–611. doi:10.1016/S0140-6736(16)32410-2
- 49 Kuramatsu JB, Gerner ST, Ziai W, et al. Association of intraventricular fibrinolysis with clinical outcomes in ICH: an individual participant data meta-analysis. *Stroke* 2022;53(9):2876–2886. doi:10.1161/STROKEAHA.121.038455
- 50 Staykov D, Huttner HB, Struffert T, et al. Intraventricular fibrinolysis and lumbar drainage for ventricular hemorrhage. *Stroke* 2009;40(10): 3275–3280. doi:10.1161/STROKEAHA.109.551945
- 51 Urdy S, Kimberly WT, Beslow LA, et al. Targeting secondary injury in intracerebral haemorrhage-perihaematoma oedema. *Nat Rev Neurol* 2015; 11(2):111–122. doi:10.1038/nrneurol.2014.264
- 52 Wu TY, Sharma G, Strbian D, et al. Natural history of perihematoma edema and impact on outcome after intracerebral hemorrhage. *Stroke* 2017;48(4):873–879. doi:10.1161/STROKEAHA.116.014416
- 53 Ironside N, Chen CJ, Ding D, Mayer SA, Connolly ES. Perihematoma edema after spontaneous intracerebral hemorrhage. *Stroke* 2019;50(6): 1626–1633. doi:10.1161/STROKEAHA.119.024965

- 54 Murthy SB, Moradiya Y, Dawson J, et al. Perihematomal edema and functional outcomes in intracerebral hemorrhage: influence of hematoma volume and location. *Stroke* 2015; 46(11):3088–3092. doi:10.1161/STROKEAHA.115.010054
- 55 Kamel H, Navi BB, Nakagawa K, Hemphill JC, Ko NU. Hypertonic saline versus mannitol for the treatment of elevated intracranial pressure: a meta-analysis of randomized clinical trials. *Crit Care Med* 2011;39(3):554–559. doi:10.1097/CCM.0b013e318206b9be
- 56 Cook AM, Morgan Jones G, Hawryluk GWJ, et al. Guidelines for the acute treatment of cerebral edema in neurocritical care patients. *Neurocrit Care* 2020;32(3):647–666. doi:10.1007/s12028-020-00959-7
- 57 Leasure AC, Qureshi AI, Murthy SB, et al. Intensive blood pressure reduction and perihematomal edema expansion in deep intracerebral hemorrhage. *Stroke* 2019;50(8):2016–2022. doi:10.1161/STROKEAHA.119.024838
- 58 Mould WA, Carhuapoma JR, Muschelli J, et al. Minimally invasive surgery plus recombinant tissue-type plasminogen activator for intracerebral hemorrhage evacuation decreases perihematomal edema. *Stroke* 2013;44(3):627–634. doi:10.1161/STROKEAHA.111.000411
- 59 Volbers B, Herrmann S, Willfarth W, et al. Impact of hypothermia initiation and duration on perihemorrhagic edema evolution after intracerebral hemorrhage. *Stroke* 2016;47(9):2249–2255. doi:10.1161/STROKEAHA.116.013486
- 60 Mendelow AD, Gregson BA, Fernandes HM, et al. Early surgery versus initial conservative treatment in patients with spontaneous supratentorial intracerebral haematomas in the International Surgical Trial in Intracerebral Haemorrhage (STICH): a randomised trial. *Lancet* 2005;365(9457):387–397. doi:10.1016/S0140-6736(05)17826-X
- 61 Mendelow AD, Gregson BA, Rowan EN, et al. Early surgery versus initial conservative treatment in patients with spontaneous supratentorial lobar intracerebral haematomas (STICH II): a randomised trial. *Lancet* 2013; 382(9890):397–408. doi:10.1016/S0140-6736(13)60986-1
- 62 Hanley DF, Thompson RE, Rosenblum M, et al. Efficacy and safety of minimally invasive surgery with thrombolysis in intracerebral haemorrhage evacuation (MISTIE III): a randomised, controlled, open-label, blinded endpoint phase 3 trial. *Lancet* 2019;393(10175):1021–1032. doi:10.1016/S0140-6736(19)30195-3
- 63 Pradilla G, Ratcliff JJ, Hall AJ, et al; ENRICH trial investigators; ENRICH Trial Investigators. Trial of early minimally invasive removal of intracerebral hemorrhage. *N Engl J Med* 2024;390(14):1277–1289. doi:10.1056/NEJMoa2308440
- 64 Kobata H, Ikeda N. Recent updates in neurosurgical interventions for spontaneous intracerebral hemorrhage: minimally invasive surgery to improve surgical performance. *Front Neurol* 2021;12:703189. doi:10.3389/fneur.2021.703189
- 65 Kuramatsu JB, Biffi A, Gerner ST, et al. Association of surgical hematoma evacuation vs conservative treatment with functional outcome in patients with cerebellar intracerebral hemorrhage. *JAMA* 2019;322(14):1392–1403. doi:10.1001/jama.2019.13014
- 66 Gross BA, Jankowitz BT, Friedlander RM. Cerebral intraparenchymal hemorrhage: a review. *JAMA* 2019;321(13):1295–1303. doi:10.1001/jama.2019.2413
- 67 Huttner HB, Schwab S. Malignant middle cerebral artery infarction: clinical characteristics, treatment strategies, and future perspectives. *Lancet Neurol* 2009;8(10):949–958. doi:10.1016/S1474-4422(09)70224-8
- 68 Schirmer CM, Ackil AA, Malek AM. Decompressive craniectomy. *Neurocrit Care* 2008;8(3):456–470. doi:10.1007/s12028-008-9082-y
- 69 Gregson BA, Broderick JP, Auer LM, et al. Individual patient data subgroup meta-analysis of surgery for spontaneous supratentorial intracerebral hemorrhage. *Stroke* 2012;43(6):1496–1504. doi:10.1161/STROKEAHA.111.640284
- 70 Marenco-Hillebrand L, Suarez-Meade P, Ruiz Garcia H, et al. Minimally invasive surgery and transsulcal parafascicular approach in the evacuation of intracerebral haemorrhage. *Stroke Vasc Neurol* 2019;5(1):40–49. doi:10.1136/svn-2019-000264
- 71 Hersh EH, Gologorsky Y, Chartrain AG, Mocco J, Kellner CP. Minimally invasive surgery for intracerebral hemorrhage. *Curr Neurol Neurosci Rep* 2018;18(6):34. doi:10.1007/s11910-018-0836-4
- 72 Scaggiante J, Zhang X, Mocco J, Kellner CP. Minimally invasive surgery for intracerebral hemorrhage. *Stroke* 2018;49(11):2612–2620. doi:10.1161/STROKEAHA.118.020688
- 73 Kellner CP, Song R, Ali M, et al. Time to evacuation and functional outcome after minimally invasive endoscopic intracerebral hemorrhage evacuation. *Stroke* 2021;52(9):e536–e539. doi:10.1161/STROKEAHA.121.034392
- 74 van Loon J, Van Calenberg F, Goffin J, Plets C. Controversies in the management of spontaneous cerebellar haemorrhage. A consecutive series of 49 cases and review of the literature. *Acta Neurochir (Wien)* 1993;122(3-4):187–193. doi:10.1007/BF01405527
- 75 De Herdt V, Dumont F, Hénon H, et al. Early seizures in intracerebral hemorrhage: incidence, associated factors, and outcome. *Neurology* 2011;77(20):1794–1800. doi:10.1212/WNL.0b013e31823648a6

- 76 Peter-Derex L, Philippeau F, Garnier P, et al. Safety and efficacy of prophylactic levetiracetam for prevention of epileptic seizures in the acute phase of intracerebral haemorrhage (PEACH): a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet Neurol* 2022;21(9):781-791. doi:10.1016/S1474-4422(22)00235-6
- 77 Merkler AE, Gialdini G, Lerario MP, et al. Population-based assessment of the long-term risk of seizures in survivors of stroke. *Stroke* 2018; 49(6):1319-1324. doi:10.1161/STROKEAHA.117.020178
- 78 Haapaniemi E, Strbian D, Rossi C, et al. The CAVE score for predicting late seizures after intracerebral hemorrhage. *Stroke* 2014;45(7): 1971-1976. doi:10.1161/STROKEAHA.114.004686
- 79 Gillow SJ, Ouyang B, Lee VH, John S. Factors associated with fever in intracerebral hemorrhage. *J Stroke Cerebrovasc Dis* 2017;26(6): 1204-1208. doi:10.1016/j.jstrokecerebrovasdis.2017.01.007
- 80 Liddle LJ, Dirks CA, Almekhlafi M, Colbourne F. An ambiguous role for fever in worsening outcome after intracerebral hemorrhage. *Transl Stroke Res* 2023;14(2):123-136. doi:10.1007/s12975-022-01010-x
- 81 Kang K, Lu J, Ju Y, et al. Association of pre- and post-stroke glycemic status with clinical outcome in spontaneous intracerebral hemorrhage. *Sci Rep* 2019;9. doi:10.1038/s41598-019-55610-z
- 82 Qureshi AI, Huang W, Lobanova I, et al. Effect of moderate and severe persistent hyperglycemia on outcomes in patients with intracerebral hemorrhage. *Stroke* 2022;53(4):1226-1234. doi:10.1161/STROKEAHA.121.034928
- 83 NICE-SUGAR Study Investigators, Finfer S, Chittock DR, et al. Intensive versus conventional glucose control in critically ill patients. *N Engl J Med* 2009;360(13):1283-1297. doi:10.1056/NEJMoa0810625
- 84 Johnston KC, Bruno A, Pauls Q, et al. Intensive vs standard treatment of hyperglycemia and functional outcome in patients with acute ischemic stroke: the SHINE Randomized Clinical Trial. *JAMA* 2019;322(4):326-335. doi:10.1001/jama.2019.9346
- 85 Chu Q, Liao L, Wei W, et al. Venous thromboembolism in ICU patients with intracerebral hemorrhage: risk factors and the prognosis after anticoagulation therapy. *Int J Gen Med* 2021;14:5397-5404. doi:10.2147/IJGM.S327676
- 86 CLEAR III Trial Investigators. Venous thromboembolism after intraventricular hemorrhage: results from the CLEAR III trial. *Neurosurgery* 2019;84(3):709-716. doi:10.1093/neuros/nyy189
- 87 Cheng X, Zhang L, Xie NC, Ma YQ, Lian YJ. High plasma levels of D-dimer are independently associated with a heightened risk of deep vein thrombosis in patients with intracerebral hemorrhage. *Mol Neurobiol* 2016;53(8): 5671-5678. doi:10.1007/s12035-015-9487-5
- 88 Li L, Murthy SB. Cardiovascular events after intracerebral hemorrhage. *Stroke* 2022;53(7): 2131-2141. doi:10.1161/STROKEAHA.122.036884
- 89 CLOTS (Clots in Legs Or sTockings after Stroke) Trials Collaboration, Dennis M, Sandercock P, et al. Effectiveness of intermittent pneumatic compression in reduction of risk of deep vein thrombosis in patients who have had a stroke (CLOTS 3): a multicentre randomised controlled trial. *Lancet* 2013;382(9891):516-524. doi:10.1016/S0140-6736(13)61050-8
- 90 Hemphill JC, Bonovich DC, Besmertis L, Manley GT, Johnston SC. The ICH score: a simple, reliable grading scale for intracerebral hemorrhage. *Stroke* 2001;32(4):891-897. doi:10.1161/01.str.32.4.891
- 91 Cheung RTF, Zou LY. Use of the original, modified, or new intracerebral hemorrhage score to predict mortality and morbidity after intracerebral hemorrhage. *Stroke* 2003;34(7):1717-1722. doi:10.1161/01.STR.0000078657.22835.B9
- 92 Godoy DA, Piñero G, Di Napoli M. Predicting mortality in spontaneous intracerebral hemorrhage: can modification to original score improve the prediction? *Stroke* 2006;37(4): 1038-1044. doi:10.1161/01.STR.0000206441.79646.49
- 93 Weimar C, Benemann J, Diener H. Development and validation of the Essen Intracerebral Haemorrhage Score. *J Neurol Neurosurg Psychiatry* 2006;77(5):601-605. doi:10.1136/jnnp.2005.081117
- 94 Rost NS, Smith EE, Chang Y, et al. Prediction of functional outcome in patients with primary intracerebral hemorrhage: the FUNC score. *Stroke* 2008;39(8):2304-2309. doi:10.1161/STROKEAHA.107.512202
- 95 Ji R, Shen H, Pan Y, et al. A novel risk score to predict 1-year functional outcome after intracerebral hemorrhage and comparison with existing scores. *Crit Care* 2013;17(6):R275. doi:10.1186/cc13130
- 96 Sembill JA, Gerner ST, Volbers B, et al. Severity assessment in maximally treated ICH patients: the max-ICH score. *Neurology* 2017;89(5): 423-431. doi:10.1212/WNL.0000000000004174
- 97 Shah VA, Thompson RE, Yenokyan G, et al. One-year outcome trajectories and factors associated with functional recovery among survivors of intracerebral and intraventricular hemorrhage with initial severe disability. *JAMA Neurol* 2022;79(9):856-868. doi:10.1001/jamaneurol.2022.1991